Xinyue (Susan) Zeng

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Research Focus

Research on reliable and efficient LLM reasoning through RL post-training, adaptive test-time scaling, and long-horizon agents. Four first-author papers accepted at NeurIPS, ICLR*2, and ICML; research internship experience at Microsoft Frontier Tuning AI.

Education

Virginia Tech, Ph.D. Candidate in Computer Science; *Advised by Prof. Dawei Zhou* 08/24 – 05/28
UNC–Chapel Hill, M.S. in Biostatistics 08/22 – 05/24
Zhejiang University, B.S. in Statistics (Honors) 08/18 – 07/22

Work Experience

Microsoft Frontier Tuning AI, *Research Intern* May 2026 – Aug 2026

- **Self-Improving Policies for Long-Horizon Agents**
  - **ForkPilot: Self-Evolving Policy for Retrospective Search in Long-Horizon Agents** [In Submission]
    - Introduced Search Value Dynamics and developed a self-improving two-stage policy that learns from completed trajectories offline and incorporates new execution outcomes into subsequent policy updates.
    - Matched competitive agent performance with **up to 59.2% fewer tokens** across 6 benchmarks and 7 LLM families, compared with 9 baselines.
    - Reduced token usage by **up to 80.9%** on 3 internal workflow benchmarks, demonstrating transfer to practical agent workloads.

Virginia Tech, *Graduate Research Assistant* Aug 2024 – Present

- **RL Post-Training and Long-Horizon Reasoning**
  - **SAGE: Mitigating Long-Horizon Reasoning Biases via Topological Guidance** [NeurIPS 2026]
    - Introduced Symbolic Closure Analysis to characterize exploration and compounding biases in long-horizon reasoning; developed algebraic sparsification and hyperbolic structural guidance to improve RL post-training.
    - Achieved **up to 8-fold higher success rates** on Andrews–Curtis reasoning tasks with **zero additional inference overhead**, evaluating 12 benchmarks and 7 model families.
- **Reliable LLM Reasoning and Evaluation**
  - **HalluGuard: Demystifying Data-Driven and Reasoning-Driven Hallucinations in LLMs** [ICLR 2026]
    - Developed and implemented an NTK-based hallucination-scoring method that jointly captures data-driven and reasoning-driven errors, enabling reference-free detection and score-guided test-time reasoning.
    - Achieved state-of-the-art hallucination detection across 10 benchmarks and 9 LLM backbones, evaluated against 11 baselines; delivered up to **28.5% relative accuracy gains** in test-time reasoning accuracy.
- **Efficient Test-Time Scaling and Model Selection**
  - **Plan and Budget: Effective and Efficient Test-Time Scaling on Reasoning Large Language Models** [ICLR 2026]
    - Built a model-agnostic PyTorch inference pipeline using query decomposition and uncertainty-aware token budgeting for adaptive test-time computation.
    - Improved reasoning accuracy by **up to 70% relative** while reducing token usage by **up to 39%**, without model retraining.
  - **LENSLLM: Unveiling Fine-Tuning Dynamics for LLM Selection** [ICML 2025]
    - Developed and implemented a rectified scaling model that predicts downstream fine-tuning performance, enabling efficient LLM selection without fully fine-tuning every candidate.
    - Achieved **91.1% relative prediction accuracy**, 0.858 Pearson correlation, and **up to 88.5% lower model-selection cost**.

Publications

[NeurIPS'26] SAGE: Mitigating Long-Horizon Reasoning Biases via Topological Guidance
Xinyue Zeng, Jiawei Zhang, Yujun Yan, Dawei Zhou

[ICLR'26] HalluGuard: Demystifying Data-Driven and Reasoning-Driven Hallucinations in LLMs
Xinyue Zeng[*], Junhong Lin[*], Yujun Yan, Feng Guo, Liang Shi, Jun Wu, Dawei Zhou

[ICLR'26] Plan and Budget: Effective and Efficient Test-Time Scaling on Reasoning Large Language Models
Xinyue Zeng[*], Junhong Lin[*], Jie Zhu, Song Wang, Julian Shun, Jun Wu, Dawei Zhou

[ICML'25] LENSLLM: Unveiling Fine-Tuning Dynamics for LLM Selection
Xinyue Zeng, Haohui Wang, Junhong Lin, Jun Wu, Tyler Cody, Dawei Zhou

[TMLR] Cognitive Episodes in LLM Reasoning Traces Enable Interpretable Human Item Difficulty Prediction
Chenguang Wang*, Ming Li*, **Xinyue Zeng**, Zhuochun Li, Hong Jiao, Tianyi Zhou, Dawei Zhou

[Genome Biology] Cell type-specific inference from bulk RNA-sequencing data by integrating single-cell reference profiles via EPIC-unmix
Chenwei Tang*, Quan Sun*, **Xinyue Zeng**, Gang Li, Xiaoyu Yang, Fei Liu, Jinying Zhao, Yin Shen, Boxiang Liu, Jia Wen, Yun Li

[In Submission] ForkPilot: Self-Evolving Policy for Retrospective Search in Long-Horizon Agents
Xinyue Zeng, Shivam Shandilya, Guilherme Potje, Leonardo Nunes, Rakshanda Agarwal, Ranveer Chandra, Emre Kiciman, Dawei Zhou, Tusher Chakraborty

[In Submission] How Can Rhetoric Reward-Hack AI Reviewers? Dissecting Rhetorical Sensitivity in AI-Based Peer Review
Ming Li*, Chenguang Wang*, **Xinyue Zeng**, Xirui Li, Dianqi Li, Peng Shi, Dawei Zhou, Tianyi Zhou

[In Submission] DISPROTBENCH: Uncovering the Functional Limits of Protein Structure Prediction Models in Intrinsically Disordered Regions
Xinyue Zeng[*], Tuo Wang[*], Adithya Kulkarni, Alexander Lu, Alexandra Ni, Phoebe Xing, Junhan Zhao, Siwei Chen, Dawei Zhou

[In Submission] DiagnoLLM: A Hybrid Bayesian Neural Language Framework for Interpretable Disease Diagnosis
Xinyue Zeng[*], Bowen Xu[*], Jiazhen Hu, Tuo Wang, Adithya Kulkarni

[In Submission] Scientific Hypothesis Generation and Validation: Methods, Datasets, and Future Directions
Adithya Kulkarni, Fatimah Alotaibi, **Xinyue Zeng**, Longfeng Wu, Tong Zeng, Barry Menglong Yao, Minqian Liu, Shuaicheng Zhang, Lifu Huang, Dawei Zhou

[*] Equal contribution.

Patent

Fork Allocation Training for Multi-turn Machine-learning Models
Microsoft patent application · Inventor · Under application review (2026).

Selected Honors

Microsoft Patent Award, Microsoft Frontier Tuning AI 2026
Tier One Financial Travel Award, ICLR 2026 *(for accepted papers)* 2026
Student Spotlight, Sanghani Center for AI & Data Analytics 2026
Graduate Student Travel Fellowship, Virginia Tech College of Engineering 2026
Travel Scholarship, ACM CAPWIC *(for invited presentation)* 2026

Selected Invited Talks

NICE AI, Hallucinations in LLMs Apr 2026
Cerebras AI, Fine-Tuning Dynamics for LLM Selection Oct 2025
Ploutos AI, Fine-Tuning Dynamics for LLM Selection Oct 2025